Decreasing Universe Theory: The Tully-Fisher Relation

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Abstract: Starting from the 'Decreasing Universe Theory' (DUT), particularly from the formula that explains the *dark matter illusion* [04], we will derive the Tully-Fisher relation [01] for spiral galaxies.

Keywords: *Tully-Fisher, Dark Energy, Hubble's Law, Decreasing Universe Theory, Universe Expansion, Dark Matter*

1 - Introduction

The "Decreasing Universe Theory" starts from the premise that the gravitational field causes space and everything contained within it to *continuously* shrink. In places where the gravitational field is zero, no shrinkage occurs.

From this premise, we have already reached the following results:

1. We derived Hubble's Law [02] (without requiring dark energy)

2. We derived a simple formula for the distance of a galaxy as a function of its redshift (when the galaxy is similar to via-lactea) [03]

3. We derived the rotation curve of galaxy Messier 33: We explained why *dark matter* is an illusion and how the stellar rotation curve in the galaxy can be obtained without invoking such an entity [04]

4. We deduced (without postulating), that the speed of light is invariant for references that are contracting due to the gravitational field and those that are not [05]

5. We derived the "*Cosmological Time Dilation*" and showed how this can be applied to the explosion time of SN1A-type Supernovae, which grows by the factor (Z+1) [05]

In this article we will derive the Tully-Fisher relation.

2 - The Tully-Fisher Relation

The Tully–Fisher relation [01] is an empirical law of astrophysics connecting two observables in spiral galaxies:

- how fast they spin
- how bright (or massive) they are

$$\mathbf{M \propto V^4} \qquad \textbf{[F00]}$$

Tully-Fisher Relation

Where:
M = Mass or luminosity (brightness of the galaxy)
V = rotation velocity

Tully-Fisher establishes that the mass of a spiral galaxy is proportional to the fourth power of its velocity (the velocity of the stars at the outermost orbits).

We will now derive the so-called 'Tully-Fisher Relation' (TFR) from the DUT.

3 - Some Formulas and Concepts

To arrive at the TFR, we will use some formulas and concepts established in previous articles [06].

First, let us recall the main formulas and results obtained.

The main formula is the one relating the measured contraction of space (and everything contained within it) to the measurement obtained by a reference frame that does not contract, and vice versa.

Thus, the distance measure (L) of sidereal space (which does not contract), observed by a reference frame **that is contracting**, is given by [02][04]:

$$\mathrm{L(t)} = \frac{\mathrm{L_0}}{2^{\Delta \mathrm{t}/T_j}} \qquad \text{(F1)}$$

(Contraction formula from the point of view of a non-shrinking observer)

Where:
$L(t)$ = Measure of L_0 (decreasing) by a "Sidereal Observer".
t_0 = Initial time (arbitrary)
L_0 = Length measured at $t=t_0$
Tj = Jocaxian Time (Time to shrink to half)
$\Delta t = t - t_0$

Rewriting and using the *Jocaxian Factor* (Fj):

$$F_j(\Delta \mathrm{t}) = 2^{\Delta \mathrm{t}/T_j} \qquad (F2)$$

Jocaxian's Factor

Or, alternatively:

:

$$F_j(\Delta t) = e^{\left(\ln(2) * \frac{\Delta t}{T_j}\right)} \qquad (\text{ F3 })$$

Thus (F1):

$$L = L_0 / F_j(\Delta t) \qquad (\text{ F4A })$$

The 'comoving' length decreases if the distance measured is in the contracting reference frame.

If the object or distance being measured is in the non-contracting reference frame, the opposite occurs: from Earth (which is contracting), we will see the object (or distance) *increase in size*, not decrease:

$$L = L_0 * F_j(\Delta t) \qquad (\text{ F4B })$$

The 'proper' length increases if the distance is in the non-contracting reference frame.

From our original article [02], we associated the 'Jocaxian Factor' with the Hubble constant:

$$D(\Delta t) = D_0 * F_j(\Delta t) \qquad (\text{ F5 })$$

(Distance Formula with the Jocaxian Factor)

Where:
"t_0" is the time at which the photon was emitted by the galaxy.
"t" is the time at which Earth received this photon.
"D_0" is the distance we would measure from Earth to the galaxy at time "t_0".
"$D(\Delta t)$" is the distance measured from Earth to a galaxy after "Δt" time.
"Δt" = $t-t_0$ (Time period)

3.1 - Hubble's Law

Differentiating with respect to 'T' we have:

$$V = d[\ D(\Delta t)\ ] / dt \qquad (\text{ F6A })$$

$$V = (\ \ln(2) / T_j\ ) * D(\Delta t) \qquad (\text{ F6B })$$

(Distant galaxies' apparent distance speed formula.)

But Hubble's Law is:

$$V = H_0 * D \qquad (\text{ F7 })$$

(Hubble's Law)

Where H0 = Hubble's Constant
D = Distance from the galaxy)

Therefore:

$$H_0 = (\ln(2) / T_j) \qquad (F8)$$

Hubble Constant in terms of Jocaxian time

3.2 - DUT Parameters and Constants

We define the Jocaxian factor [04] as:

$$F_j(\Delta t) = e^{\left(\ln(2) * \frac{\Delta t}{T_j} \right)} \qquad (F9)$$

Or in terms of the Hubble constant:

$$F_j(\Delta t) = \exp(H_0 * \Delta t) \qquad (F10)$$

(Jocaxian Factor in terms of the Hubble constant)

3.3 - A New Hypothesis

To adapt the formula for use in another galaxy (or another location) with a different gravitational field from where we are — that is, to adapt formula [F10] for a reference frame with gravitational field **'g'** (different from our position in the Milky Way = **'g_t'**) — we made an additional hypothesis:

"The contraction time is inversely proportional to the intensity of the gravitational field." [04]

Thus, for the contraction time to halve:

$$T_j(g) = T_j \cdot \frac{g_t}{g} \qquad (F11)$$

Where:
*T_j=Jocaxian Time (**Time to Length contract to half**)*
$g_t = GMt/R_t^2$ (gravitational field at Earth (Milk Way)).
$g = GM/R^2$ (gravitational field at distance R from galactic center).

This leads us to the **generalized Jocaxian Factor**:

$$F_j(\Delta t, M, R) = e^{\frac{J_0 M \Delta t}{R^2}} \qquad (F12)$$

Where:

$$J_0 = \frac{H_0 R_t^2}{M_t} \quad \textbf{(F13)}$$

(***Jocax Constant*** ≈ ***1,37E-18*** *$m^2 \cdot s^{-1} \cdot kg^{-1}$*).

Where
H_0=Hubble Constant (≈2,2E-18 s^{-1})
R_t = Distance from Earth to center of Milk Way (≈2,5E20 m)
M_t= Mass of Milk Way (≈1E41 Kg)
Where:
H_0 = Hubble Constant (≈2.2E-18 s^{-1})
Rt = Distance from Earth to center of Milky Way (≈2.5E20 m)
Mt = Mass of Milky Way (≈1E41 kg)

With this new hypothesis, we were able to write [F4A] in generic form. Thus, the measurement observed in the non-contracting (comoving) reference frame of something contracting in an arbitrary field 'g' is:

$$\mathbf{L(t) = L_0/F_j(M,R,\Delta T)} \quad \textbf{[F14]}$$

comoving measurement in a field 'g' different from Earth's (gt)

4 - The Dark Matter Equation

Let us now recall the derivation of the Dark Matter equation [04] to understand the process:

To understand what happens to the photon until it reaches Earth, we must consider that:

a) *Both the galaxy where the photon originated and the Milky Way have been contracting since their formation. We will assume both began contracting simultaneously at instant T_0. (We will treat T_0 as the time from galaxy formation to the present: approximately 13 billion years = 4E16 s).*

b) *The contraction rate, as we have seen, depends on the mass of the galaxy and therefore can vary between galaxies. The generalized Jocaxian Factor [F12] accounts for these differences.*

c) *It is important to note, as seen previously, that the contraction rate of the star emitting the photon depends on its distance (R) from the galactic center. Thus, the redshift of the emitted photon (considering only this factor) will be greater the farther the star is from the galactic center.*

d) *Obviously, part of the redshift is due to the rotation velocity of the star, which is greater the closer the star is to the center.*

e) *This difference in the outgoing redshift, as a function of distance from the galactic center, will define the "dark matter effect" and its peculiar curve.*

f) *Consider that T_1 is the instant when the photon is emitted from the source galaxy toward Earth. When it leaves the source galaxy, it stops contracting, since the galaxy's gravitational field becomes negligible.*

g) *We will calculate the apparent velocity of the star (as a function of its distance from the center), considering the redshift until it leaves the galaxy. (We will not consider the 'dark energy' effect, which is the contraction of our galaxy after the photon leaves the emitting galaxy until it arrives at Earth).*

4.1 - The Equations

For simplicity, let us assume that the galaxies formed at the same time. In that case, photons began contracting differently in each galaxy with different masses.

The total redshift is the change in wavelength relative to a standard on Earth.

Initially, shortly after their formation, both the emitting galaxy and the receiving galaxy (the Milky Way) emitted photons at the same wavelength (=$\boldsymbol{\lambda_0}$).

After about 13 billion years (=$\mathbf{T_0}$), the contraction rates differed, and therefore the wavelength decreased differently for both galaxies.

In the Milky Way, the photon after $\mathbf{T_0}$ will have a shorter wavelength:

$$\boldsymbol{\lambda^T{}_0 = \lambda_0/F^T{}_J(T_0) = \lambda_0/\exp(H_0T_0)} \qquad \text{[F15]}$$

Where:
λ_0 = Wavelength at beginning
$\lambda(T_0)$ = Wavelength of photon on Earth after T_0
FTJ = Jocaxian Factor on Earth
T_0 = Time from beginning of Galaxy until Now (13G Years)

In the emitting galaxy, *the contraction rate is different*, so we need to use the general formula [F14]:

$$\lambda^{M}_{0} = \lambda_0 / F_j(T_0, M, R) \quad [F16]$$

Where:
$\lambda(M_0)$ = Wavelength of photon in another galaxy after T_0
$Fj(T_0, M, R)$ = Generalized Jocaxian Factor
M = Mass of Galaxy
R = Distance of star to center of galaxy

However, the star emitting the photon will have a redshift that depends on its rotation velocity:

$$\mathbf{V = SQRT(MG/R)} \quad \mathbf{[F17]}$$

Recalling that:

$$\mathbf{V = cz} \quad \mathbf{[F18]}$$

Where: c = Speed of light
z = redshift

$$\mathbf{z = V/c = (\lambda(M_1) / \lambda(M_0) - 1)} \quad \mathbf{[F19]}$$

Where:
$\lambda(M_1)$ = Wavelength of photon with galaxy rotation velocity

From [F17][F18][F19] we derive:

$$\mathbf{\lambda(M_1) = \lambda(M_0)\ (SQRT(MG/R)/c + 1)} \quad \mathbf{[F20]}$$

The final redshift relative to the Milky Way will be:

$$\mathbf{Zf = (\lambda(M_1) - \lambda(T_0)) / \lambda(T_0)} \quad \mathbf{[F21]}$$

Thus:

$$\mathbf{Zf = (\lambda(M_0)\ (SQRT(MG/R)/c + 1)) / \lambda(T_0) - 1} \quad \mathbf{[F22]}$$

Substituting [F15] and [F16]:

$$\mathbf{Zf = (exp(H_0T_0) / Fj(T_0,M,R))\ (SQRT(MG/R)/c + 1) - 1} \quad \mathbf{[F23]}$$

But v = cz, then the speed measured on Earth (without considering the effect of 'dark energy' = our contraction during the photon's journey) will be:

$$\mathbf{V = c\ \{ (exp(H_0T_0) / Fj(T_0,M,R))\ (SQRT(MG/R)/c + 1) - 1 \}} \quad \mathbf{[F24]}$$

Substituting Fj(T_0,M,R), we obtain the dark matter equation — the apparent rotation curve of the galaxy:

$$\mathbf{V=c\ (}\left(\frac{1}{c}\sqrt{\frac{MG}{R}}+1\right)e^{H_0T_0-\frac{J_0MT_0}{R^2}}\ \mathbf{-1)}\quad \mathbf{[F25]}$$

Galaxy Rotation Curve ('Dark Matter Equation')

Where:
C = Speed of Light (3E8 m/s)
M = Mass of the Emitting Galaxy (1.4E40 kg)
G = Gravitational Constant (6.7E-11)
R = Distance from the Star to the center of the galaxy
H_0 = Hubble Constant (2.2E-18)
T_0 = Contraction Time (13 billion years = 4E16s)
J_0 = Jocax Constant (1.37E-18)

5 - Brightness is Proportional to Density

For spiral galaxies dominated by baryonic disks, the observed luminous distribution is frequently used as a direct tracer of the stellar surface density, especially when one assumes an approximately constant mass-to-light ratio *((M/L))* in red or infrared optical bands.

In this context, the stellar surface density can be written as: Σ(R) = (M/L)I(R)

where Σ(R) represents the baryonic/stellar surface density and I(R) is the surface brightness.

Thus, greater luminosity per unit area implies a greater concentration of visible baryonic mass. *This approximation is widely employed in structural studies of spiral disks*. (Aanda)

$$\mathbf{M/L \approx Cte_1} \qquad \mathbf{[F26]}$$

The Mass/Luminosity ratio is constant in spiral galaxies

Where:
M = Mass of the galaxy
L = Luminosity of the galaxy

Historically, Freeman (1970) observed that many high-surface-brightness spiral galaxies had relatively similar central surface brightness values, a phenomenon known as **Freeman's Law** **[07]**, suggesting that a large fraction of these galaxies share a relatively narrow range of central surface brightness. Although subsequent work has shown greater scatter due to the inclusion of low-surface-brightness galaxies, the relation remains relevant for classical high-brightness spirals. (OUP Academic)

$$\mathbf{L/A \approx Cte_2} \qquad \mathbf{[F27]}$$

Under the hypothesis that ***surface brightness approximately traces baryonic surface density***, the observational similarity of central surface brightness implies that many classical spiral galaxies have relatively similar average baryonic mass densities, at least as a first-order approximation.

In other words, if surface luminosity and baryonic density are correlated, then spiral disks with similar surface brightness will also tend to show comparable average surface mass distributions.

This regularity provides an observational basis for simplified models in which ***spiral galaxies share similar baryonic structural properties***.

Thus from [F26] and [F27] we have (for a significant subset of high-surface-brightness spirals):

$$\boldsymbol{\rho \equiv M/A \approx Cte} \qquad \textbf{[F28]}$$

The surface mass density for spiral galaxies are very similar

Where:
ρ = Surface mass density
M = Mass of the galaxy
A = Area of the Galaxy ($\pi R^2 max$)

6 - Deriving Tully-Fisher

From the dark matter equation [F25] and under the simplifying hypotheses [F26] and [F27], we can derive the Tully-Fisher relation.

We know that for a *spiral galaxy*, the mass is inscribed approximately in a circle, and therefore *this mass must be proportional to the area of that circle*:

$$\boldsymbol{M = \rho \cdot \pi \cdot R^2} \qquad \textbf{[F29]}$$

The mass of a spiral galaxy is proportional to its area

Where:
M = Mass of the galaxy
ρ = Average surface mass density
R = Radius of the galaxy

The Milky Way is a spiral galaxy, so we can write:

$$\boldsymbol{M_v = \rho_v \cdot \pi \cdot R_v^2} \qquad \textbf{[F30]}$$

Mass as a function of radius for the Milky Way

We can now rewrite the Jocaxian constant (J_0) [F13] as a function of surface density [F30]:

$$J_0 = \frac{H_0 R_t^2}{M_t} = \frac{H_0}{\rho_t * \pi} \qquad \textbf{(F31)}$$

J_0 as a function of mass density for flat galaxies

Where:
J_0 = Jocaxian constant

H_0 = Hubble Constant
ρt = Average surface mass density of the Milky Way

We can rewrite equation [F25] in terms of surface mass densities:

$$\mathbf{V{=}c\ (}\left(\frac{1}{c}\sqrt{\frac{MG}{R}}+1\right)e^{H_0 T_0 *(1-\frac{\rho}{\rho t})}\ \mathbf{-1)}\quad \mathbf{[F32]}$$

Rotation curve as a function of surface mass density

We can use [F28] and write:

$$\rho \approx \rho t \qquad [F33]$$

Using [F33], equation [F32] becomes:

$$\mathbf{V{=}c\ (}\left(\frac{1}{c}\sqrt{\frac{MG}{R}}+1\right)e^{0}\ \mathbf{-1)} \qquad \mathbf{[F34]}$$

Using [F29]:

$$\mathbf{V=}\left(\sqrt{\frac{MG}{\sqrt{M/(\pi\rho)}}}\right) = (\sqrt{G\sqrt{\pi\rho}}\ )^{*}\ \mathbf{M^{1/4}} \qquad \mathbf{[F35]}$$

And finally:

$$\mathbf{M \propto V^4} \qquad \mathbf{[F36]}$$

Tully-Fisher relation

7 - Conclusion

In this article, we derived the Tully-Fisher relation ($M \propto V^4$) from the principles of the Decreasing Universe Theory (DUT), without invoking dark matter or any exotic componente.
.
The derivation relied on three ingredients: the DUT's galaxy rotation curve equation [F25], which emerges from the premise that gravitational fields cause space and matter to contract; and two well-established astrophysical approximations for spiral galaxies — the near-constant mass-to-light ratio [F26] and Freeman's Law of approximately uniform central surface brightness [F27].

Together, these lead naturally to a constant surface mass density across spiral galaxies [F28], which is the key step that unlocks the Tully-Fisher scaling.
This result adds to a growing list of empirical laws and observations that the DUT can reproduce from first principles — including Hubble's Law, galactic rotation curves, cosmological redshift distances, the invariance of the speed of light between contracting and non-contracting frames, and cosmological time dilation in SN1A supernovae.

The fact that a single contracting-universe premise (making it compatible with Occam's razor) can account for phenomena typically attributed to dark matter, dark energy, and relativistic effects suggests that the DUT deserves further theoretical development and observational scrutiny.

8 - References

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